Evaluation results

Generated by `python -m evaluation.run` on `cnamuangtoun/resume-job-description-fit`.

- Test split: 1759 resume-job pairs (857 No Fit, 444 Potential Fit, 458 Good Fit) across 71 jobs, 28 of which have both Good Fit and No Fit candidates and so count towards the within-job AUC.
- Training split: 6241 pairs across 280 jobs. No test job appears in training; 476 of the 477 test resumes do.
- Embedding model: `all-MiniLM-L6-v2`.

AUC is the chance that a Good Fit pair outscores a No Fit pair: 0.5 is chance, 1.0 is perfect. Brackets are 95% bootstrap confidence intervals (jobs are resampled for the within-job AUC, pairs for the other two).

Test results

Method	AUC within each job	AUC over all pairs	Spearman
TF-IDF cosine (lexical baseline)	0.682 (0.607 to 0.753)	0.581 (0.551 to 0.611)	0.128 (0.085 to 0.170)
v1: whole-document SBERT cosine	0.707 (0.634 to 0.774)	0.555 (0.523 to 0.583)	0.107 (0.062 to 0.148)
v2 component: skills match	0.654 (0.585 to 0.725)	0.501 (0.471 to 0.535)	0.017 (-0.027 to 0.066)
v2 component: requirement coverage	0.666 (0.597 to 0.735)	0.505 (0.474 to 0.541)	0.029 (-0.016 to 0.079)
v2 component: experience relevance	0.670 (0.607 to 0.735)	0.590 (0.560 to 0.621)	0.145 (0.103 to 0.189)
v2 component: education	0.507 (0.464 to 0.557)	0.466 (0.434 to 0.500)	-0.046 (-0.092 to 0.006)
v2 overall (hand-set weights)	0.674 (0.606 to 0.742)	0.515 (0.485 to 0.548)	0.041 (-0.003 to 0.089)
Leakage check: resume's training labels, job ignored	0.687 (0.606 to 0.757)	0.697 (0.667 to 0.726)	0.291 (0.245 to 0.334)
Logistic regression on the 4 v2 components	0.665 (0.602 to 0.731)	0.564 (0.532 to 0.594)	0.114 (0.068 to 0.158)
Logistic regression on interaction features	0.740 (0.657 to 0.814)	0.606 (0.577 to 0.638)	0.171 (0.128 to 0.216)
Gradient boosting on interaction features	0.731 (0.669 to 0.793)	0.648 (0.618 to 0.677)	0.218 (0.175 to 0.261)

Method	AUC within each job	AUC over all pairs	Spearman
Pairwise ranker on interaction features	0.718 (0.646 to 0.787)	0.608 (0.578 to 0.640)	0.172 (0.129 to 0.217)
Pairwise ranker on pooled embeddings	0.687 (0.597 to 0.774)	0.641 (0.610 to 0.670)	0.225 (0.179 to 0.269)
Ablation: the same without resume-only features	0.737 (0.659 to 0.809)	0.600 (0.571 to 0.632)	0.164 (0.120 to 0.209)
Fine-tuned bi-encoder	0.808 (0.735 to 0.879)	0.706 (0.679 to 0.732)	0.329 (0.288 to 0.368)

Model selection (training split only)

Within-job AUC from 5-fold cross-validation over jobs: each training pair is scored by a model that never saw its job. Methods that need no training are scored directly. The test split played no part in choosing a method.

Method	Cross-validated AUC within each job
Logistic regression on interaction features **(selected)**	0.752
Pairwise ranker on pooled embeddings	0.752
Gradient boosting on interaction features	0.749
Ablation: the same without resume-only features	0.748
TF-IDF cosine (lexical baseline)	0.732
Pairwise ranker on interaction features	0.730
v1: whole-document SBERT cosine	0.718
Logistic regression on the 4 v2 components	0.709
v2 component: requirement coverage	0.705
v2 component: experience relevance	0.700
v2 overall (hand-set weights)	0.681
Leakage check: resume's training labels, job ignored	0.651
v2 component: skills match	0.646
v2 component: education	0.535

Differences on the test split

An interval that excludes zero means the difference is unlikely to be noise.

Comparison	AUC within each job	AUC over all pairs	Spearman
v2 overall (hand-set weights) − v1: whole-document SBERT cosine	-0.033 (-0.081 to +0.012)	-0.039 (-0.072 to -0.004)	-0.066 (-0.112 to -0.014)
v2 overall (hand-set weights) − TF-IDF cosine (lexical baseline)	-0.008 (-0.062 to +0.044)	-0.066 (-0.095 to -0.036)	-0.086 (-0.131 to -0.043)
Logistic regression on interaction features − v1: whole-document SBERT cosine	+0.033 (+0.006 to +0.062)	+0.052 (+0.033 to +0.072)	+0.065 (+0.038 to +0.095)
Logistic regression on interaction features − TF-IDF cosine (lexical baseline)	+0.058 (+0.012 to +0.107)	+0.025 (-0.002 to +0.052)	+0.044 (+0.005 to +0.081)
Ablation: the same without resume-only features − v1: whole-document SBERT cosine	+0.030 (+0.005 to +0.058)	+0.045 (+0.026 to +0.066)	+0.057 (+0.030 to +0.087)
Ablation: the same without resume-only features − TF-IDF cosine (lexical baseline)	+0.055 (+0.012 to +0.101)	+0.019 (-0.008 to +0.045)	+0.037 (-0.003 to +0.073)
Fine-tuned bi-encoder − v1: whole-document SBERT cosine	+0.102 (+0.042 to +0.158)	+0.151 (+0.119 to +0.179)	+0.222 (+0.178 to +0.263)
Fine-tuned bi-encoder − TF-IDF cosine (lexical baseline)	+0.127 (+0.048 to +0.194)	+0.125 (+0.092 to +0.157)	+0.202 (+0.153 to +0.248)

Unseen-resume protocol

476 of the 477 test resumes also appear in the training split, paired with other jobs. The leakage check in the first table shows why that matters: ranking candidates purely by how their resume was labelled in training, without reading the job, already scores well. A trained model can therefore gain from remembering resumes, which would not carry over to new candidates.

Here resumes are divided into 4 folds, and each test pair is scored by a model trained without any resume from that pair's fold, so the model has seen neither the job nor the resume. Candidates are compared only with others in the same job and fold (84 such groups have both Good Fit and No Fit candidates), so these numbers are not directly comparable with the first table. Methods that need no training are unaffected by the protocol and are re-measured on the same groups.

Method	AUC within each job and fold
TF-IDF cosine (lexical baseline)	0.606 (0.539 to 0.676)
v1: whole-document SBERT cosine	0.641 (0.584 to 0.702)

Method	AUC within each job and fold
v2 overall (hand-set weights)	0.632 (0.566 to 0.703)
Logistic regression on interaction features	0.668 (0.598 to 0.741)
Ablation: the same without resume-only features	0.667 (0.599 to 0.740)
Fine-tuned bi-encoder	0.698 (0.627 to 0.772)

Comparison	Difference
v2 overall (hand-set weights) − v1: whole-document SBERT cosine	-0.009 (-0.067 to +0.052)
v2 overall (hand-set weights) − TF-IDF cosine (lexical baseline)	+0.027 (-0.028 to +0.091)
Logistic regression on interaction features − v1: whole-document SBERT cosine	+0.026 (-0.006 to +0.064)
Logistic regression on interaction features − TF-IDF cosine (lexical baseline)	+0.062 (+0.001 to +0.127)
Ablation: the same without resume-only features − v1: whole-document SBERT cosine	+0.026 (-0.004 to +0.062)
Ablation: the same without resume-only features − TF-IDF cosine (lexical baseline)	+0.061 (+0.005 to +0.122)
Fine-tuned bi-encoder − v1: whole-document SBERT cosine	+0.057 (+0.005 to +0.105)
Fine-tuned bi-encoder − TF-IDF cosine (lexical baseline)	+0.093 (+0.018 to +0.172)

Details

- Share of test pairs where each v2 component could be assessed: skills 95%, coverage 100%, experience 98%, education 59%.
- Largest weights of the pairwise ranker on interaction features (standardised inputs): `statement_mean` +0.37, `pooled_cosine` +0.27, `similarity_mean` -0.22, `coverage_min` -0.20, `coverage_weak_share` -0.18, `coverage_median` +0.17.